STA314, Lecture 5

October 26, 2021
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Recap

Goal: select relevant predictors among many candidate predictors.

We discussed best subset, forward stepwise and backward stepwise selection.

- Given p possible predictors, all three procedures produce candidate models $\mathcal{M}_0, ..., \mathcal{M}_p$ with $0, 1, ..., p$ predictors.
- Procedures differ in the way that candidate models are selected and in resulting computational cost.
- Best subset selection considers 2^p models, which is infeasible for large p. Forward and backward stepwise selection only considers $1 + p(p+1)/2$ models but is not guaranteed to find best model for a given number of predictors.

Now we need to select among those candidate models. This can not be done using RSS (why?).

One way out is cross-validation.

An alternative are analytic corrections that we discuss later today.

The right way to do cross-validation for model selection

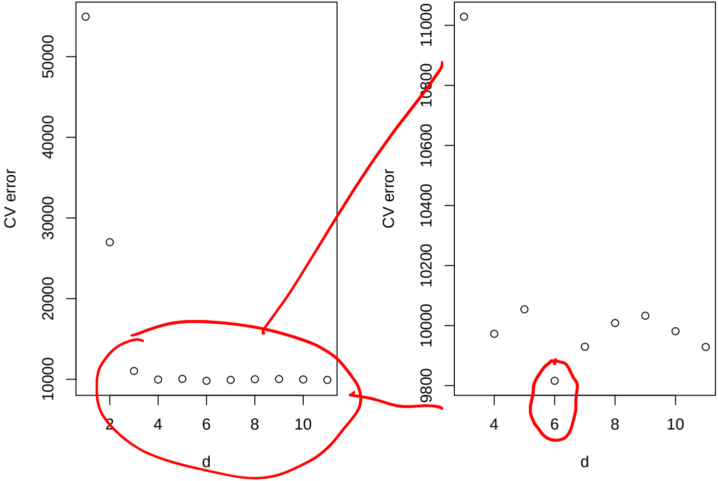
For performing L-fold cross-validation, split data into L roughly equal folds. Repeat the following steps for $\ell = 1, \ldots, L$

1. Run step 1 and step 2 of any of the selection procedures described earlier using the data not contained in fold ℓ to obtain models $\mathcal{M}_0, \ldots, \mathcal{M}_p$.
2. Use each of the models $\mathcal{M}_0, \ldots, \mathcal{M}_p$ to compute prediction errors on the data in the ℓ'th fold.

For each number of predictors $0, \ldots, p$ the above procedure gives a prediction error within each fold. Average those errors (over folds for each d) and select the number of predictors which gives the smallest error.

- Intuition: the number of predictors corresponds to a 'tuning parameter'. Models with same number of predictors but different sets of predictors have similar complexity (can overfit data to a similar degree) and thus same tuning parameter.
- Wrong way: determining candidate models $\mathcal{M}_0, \ldots, \mathcal{M}_p$ on full data set and using cross-validation afterwards to choose between models (why?).

10-fold cross-validation for predicting `Balance` in Credit data set



Which number of predictors d would cross-validation choose? $d = 6$

The general one standard error rule

Assume we have collection of estimators $\widehat{f}_d$ where $d \in \mathcal{D}$ denotes a tuning parameter.

1. $\widehat{f}_d$ for k-nn: d corresponds to k.
2. Linear models with different number of predictors: d is the number of predictors.

Assume that for each $d \in \mathcal{D}$, L-fold cross-validation returns estimated errors on each of the L folds. Call those errors $e_1(d), ..., e_L(d)$.

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1. Find d_0 which minimizes $L^{-1} \sum_{\ell=1}^{L} e_\ell(d)$ (classical cross validation would output this as solution).
2. Compute $\widehat{sd}(d_0)$, the sample standard deviation of $e_1(d_0), ..., e_L(d_0)$.
3. Among all d that satisfy

$$L^{-1} \sum_{\ell=1}^{L} e_\ell(d) \leq L^{-1} \sum_{\ell=1}^{L} e_\ell(d_0) + L^{-1/2} \widehat{sd}(d_0)$$

select the d that corresponds to the least flexible model.

The general one standard error rule II

1. Find d_0 which minimizes $L^{-1}\sum_{\ell=1}^{L} e_\ell(d)$ (classical cross validation would output this as solution).

2. Compute $\widehat{sd}(d_0)$, the sample standard deviation of $e_1(d_0), \ldots, e_L(d_0)$.

3. Among all d that satisfy

note: d_0 always satisfies

$$L^{-1}\sum_{\ell=1}^{L} e_\ell(\underset{d_0}{d}) \le L^{-1}\sum_{\ell=1}^{L} e_\ell(d_0) + \underbrace{L^{-1/2}\widehat{sd}(d_0)}_{\geq 0}$$

select the d **that corresponds to the least flexible model**.

- k-nn: larger k correspond to less flexible models, so we select the largest k.
- Linear models with different number of predictors: smaller number of predictors correspond to less flexible models, so we select model with smallest number of predictors.
- This general rule can be used whenever we prefer *less flexible* models that still have a good predictive performance. Will see more examples later.

Analytic corrections for training error I

- So far we discussed how to use cross-validation to obtain estimators of the test error. We have also seen (in earlier lectures) that for linear models R^2 is not helpful for model selection (recall why?).

- For linear models, there are other popular approaches which are based on analytic corrections. Those can be faster than cross-validation and it is important to know their meaning.

Mathematical motivation: let predictors $X_i = (1, x_{i,1}, \ldots, x_{i,d})^\top$ be $d+1$-dimensional.

- Consider n data (X_i, y_i) with $y_i = X_i^\top b + \varepsilon_i$, ε_i i.i.d. with $\mathbb{E}[\varepsilon_i] = 0, \mathbb{E}[\varepsilon_i^2] = \sigma^2$ and $\mathrm{X}^\top\mathrm{X}$ of full rank.
- Let $\hat{y}_i = X_i^\top \hat{b}$ lin. regr. estimator
- Consider $y_i^{new} = X_i^\top b + \varepsilon_i^{new}$ with ε_i^{new} i.i.d. $\mathbb{E}[\varepsilon_i^{new}] = 0, \mathbb{E}[(\varepsilon_i^{new})^2] = \sigma^2$ independent of $\varepsilon_1, \ldots, \varepsilon_n$.

One can prove (see derivation in class)

$$\underbrace{\sum_i \mathbb{E}[(y_i^{new} - \hat{y}_i)^2]}_{\mathbb{E}[\text{test error}]} - \overbrace{\sum_i \mathbb{E}[(y_i - \hat{y}_i)^2]}^{\mathbb{E}[\text{training error}]} = \boxed{2(d+1)\sigma^2}.$$

$n\sigma^2$ AIC

Interpretation: on average training error underestimates test error by $2(d+1)\sigma^2$.

Analytic corrections for training error II

Most popular approaches for least squares linear regression

training error

correction

$$\text{Mallow's } C_p: \quad C_p := \frac{1}{n}(RSS(\mathcal{M}_d) + 2(d+1)\hat{\sigma}^2)$$

$$\text{Akaike information criterion}: \quad AIC := \frac{1}{n\hat{\sigma}^2}(RSS(\mathcal{M}_d) + 2(d+1)\hat{\sigma}^2)$$

$$\text{Bayesian information criterion}: \quad BIC := \frac{1}{n\hat{\sigma}^2}(RSS(\mathcal{M}_d) + (d+1)(\log n)\hat{\sigma}^2)$$

$$\text{Adjusted } R^2: \quad aR^2 := 1 - \frac{RSS(\mathcal{M}_d)/(n-d-1)}{TSS/(n-1)}$$

natural log

$n \hat{=}$ sample size

- In each case $\hat{\sigma}^2$ denotes an estimator of the error variance $Var(\varepsilon_1)$. Usually variance estimator from 'full model' (i.e. including all predictors).
- AIC scores can be compared for models with different numbers of predictors. Select model with smallest AIC score, same for BIC and C_p

AIC is approximation to test error

- Select model with *largest* adjusted R^2.

Analytic corrections for training error III

Most popular approaches for least squares linear regression

$d \uparrow$

$$\text{Mallow's } C_p: \quad C_p := \frac{1}{n}\left(RSS(\mathcal{M}_d) + 2(d+1)\hat{\sigma}^2\right)$$

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$$\text{Bayesian information criterion}: \quad BIC := \frac{1}{n\hat{\sigma}^2}\left(RSS(\mathcal{M}_d) + (d+1)(\log n)\hat{\sigma}^2\right)$$

$$\text{Adjusted } R^2: \quad aR^2 := 1 - \frac{RSS(\mathcal{M}_d)/(n-d-1)}{TSS/(n-1)}$$

- In each case $\hat{\sigma}^2$ denotes an estimator of the error variance $Var(\varepsilon_1)$. Usually variance estimator from 'full model' (i.e. including all predictors).
- Note: in textbook C_p, AIC, BIC are given with d instead of $d+1$. This does not matter for model selection discussed next.
- C_p, AIC, BIC introduce additional 'penalty' term. Including additional predictors will decrease RSS but increase penalty term.
- Adjusted R^2 smaller than 'usual' R^2. No real theoretical justification, so should not be used in practice.

R examples: file `STA314-Lecture05-AIC-BIC-CreditAnalysis`

Comments

- Historically, AIC and BIC were derived for general *maximum likelihood* methods. Least squares regression is just one special example.

- Depending on book, software etc. AIC and BIC have different forms. All forms lead to the same model selection.

- In the setting discussed here, AIC and C_p select exactly the same model (why is this true mathematically?). Deeper reason: maximum likelihood for Gaussian errors is the least squares estimator that we considered in lectures.

$\log n > 2$

Some facts and statistical folklore: AIC vs BIC

- Fact: for $n \geq 8$ BIC corresponds to larger penalties compared to AIC (why?). Thus BIC tends to prefer smaller models compared to AIC.

- Fact: AIC does usually not lead to 'consistent model selection', it includes noise variables with high probability. BIC can perform consistent model selection (this can be formalized in a mathematical framework).

- Folklore (often true): AIC leads to models with better prediction performance. This is especially true when there are some 'weak' predictors which BIC can miss.

AIC vs BIC and consistent model selection I

Example illustrating why BIC can perform consistent model selection and AIC in general won't:

- true model $y_i = \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$. x_i fixed (non-random) one-dimensional predictors unrelated to y_i.
- Use AIC and BIC to select between model with no predictor and model with one-dimensional x_i (unrelated to y_i) as predictors.
- denote by $RSS(\mathcal{M}_0)$ residual sum of squares for model with just intercept and by $RSS(\mathcal{M}_1)$ residual sum of squares for model including x_i.
- Assume that we plug in the true value for $\sigma^2 = 1$ in AIC and BIC.

For simplicity assume $\overline{x} = n^{-1}\sum_i x_i = 0$, $\overline{x^2} = n^{-1}\sum_i x_i^2 = 0$. Then

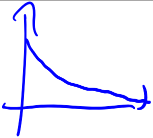
$$\hat{b}_1 = \bar{y}, \quad \hat{b}_2 = \frac{\overline{xy}}{\overline{x^2}} = \frac{n^{-1}\sum_{i=1}^n x_i y_i}{n^{-1}\sum_{i=1}^n x_i^2}$$

This implies (see derivation in class) $RSS(\mathcal{M}_0) - RSS(\mathcal{M}_1) \sim \chi_1^2$ i.e. $RSS(\mathcal{M}_0) - RSS(\mathcal{M}_1)$ follows a chi-squared distribution with one degree of freedom.

AIC vs BIC and consistent model selection II

From previous slide: $RSS(\mathcal{M}_0) - RSS(\mathcal{M}_1) \sim \chi_1^2$.

AIC vs BIC and consistent model selection II



From previous slide: $RSS(\mathcal{M}_0) - RSS(\mathcal{M}_1) \sim \chi^2_1$.

$n\hat{\sigma}^2$

AIC selects model correct $\mathcal{M}_0$ if and only if

$AIC(\mathcal{M}_1)\, n\hat{\sigma}^2$ $\quad \sigma^2 = 1$

$$AIC(\mathcal{M}_0) = RSS(\mathcal{M}_0) + 2\sigma^2 < RSS(\mathcal{M}_1) + 4\sigma^2 = \quad \Leftrightarrow \quad \chi^2_1 < 2$$

The probability of this is roughly 0.84. Otherwise the 'wrong' model is selected.

AIC vs BIC and consistent model selection II

From previous slide: $RSS(\mathcal{M}_0) - RSS(\mathcal{M}_1) \sim \chi_1^2$.

AIC selects model correct $\mathcal{M}_0$ if and only if

$$RSS(\mathcal{M}_0) + 2\sigma^2 < RSS(\mathcal{M}_1) + 4\sigma^2 \qquad \Leftrightarrow \qquad \chi_1^2 < 2$$

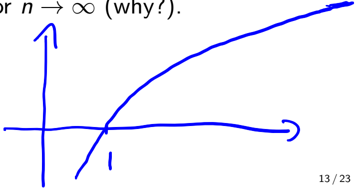
The probability of this is roughly 0.84. Otherwise the 'wrong' model is selected.

$\log n \to \infty$

BIC selects model correct $\mathcal{M}_0$ if and only if

$$RSS(\mathcal{M}_0) + \sigma^2 \log n < RSS(\mathcal{M}_1) + 2\sigma^2 \log n \qquad \Leftrightarrow \quad P(\chi_1^2 < \log n.) \to 1$$

The probability of this is depends on n but goes to 1 for $n \to \infty$ (why?).



Exercise: verify this by simulations in **R**.

Summary of results so far

We discussed best subset, forward stepwise and backward stepwise selection.

- Given p possible predictors, all three procedures produce candidate models $\mathcal{M}_0, ..., \mathcal{M}_p$ with $0, 1, ..., p$ predictors.
- Procedures differ in the way that candidate models are selected and in resulting computational cost.
- Best subset selection considers 2^p models, which is infeasible for large p. Forward and backward stepwise selection only considers $1 + p(p+1)/2$ models but is not guaranteed to find best model for a given number of predictors.

The last step of each procedure is to compare models $\mathcal{M}_0, ..., \mathcal{M}_p$ while taking into account their complexity (number of predictors). We discussed several 'analytic corrections' for this comparison.

- Basic idea of corrections: account for smaller training error in more flexible models by adding a *penalty term* that increases with number of predictors (i.e. with model flexibility).